Cheap Eyes on the Water

An open coastal temperature and tide sensor anyone can build for about forty dollars.

Rowan Hale · Design · defended on the sand at the Cove before faculty, two outside examiners, and the five people who use it

The station on the pier costs $4,000. There is one.

The coast is 70 miles long. Scientists, lifeguards, teachers, and people who fish all want to know the water, and the data comes from a handful of points. Every other station is a budget line nobody wins.

The question from Day 4 of CAP 401: what would it take to make the number cheap enough that anyone who wanted one could have one?

$41.60, from parts you can buy today

microcontroller (LoRa)	$ 12.40
pressure/temperature sensor	$ 9.80
LiFePO4 cell + charger	$ 8.20
housing (PVC, cast end caps)	$ 5.10
cable gland, bracket, strap	$ 3.60
solar 1 W	$ 2.50
total	**$ 41.60**

The build guide is 11 steps and 40 minutes. It assumes you have never soldered.

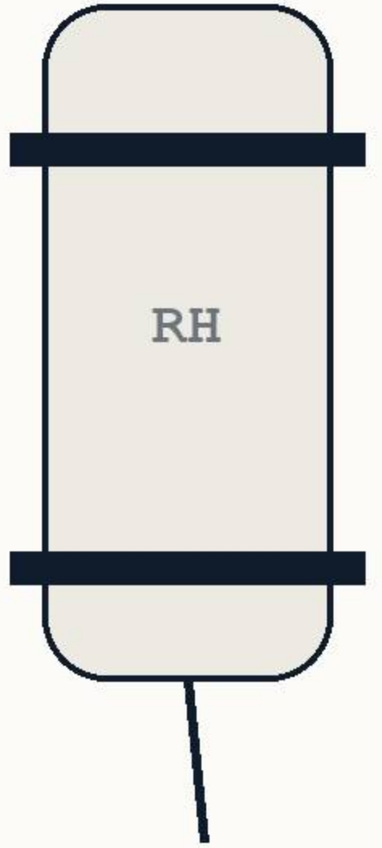
Every lesson from the pier, in PVC

Straight pull on the cable, no bends (FLD 301, week 4).

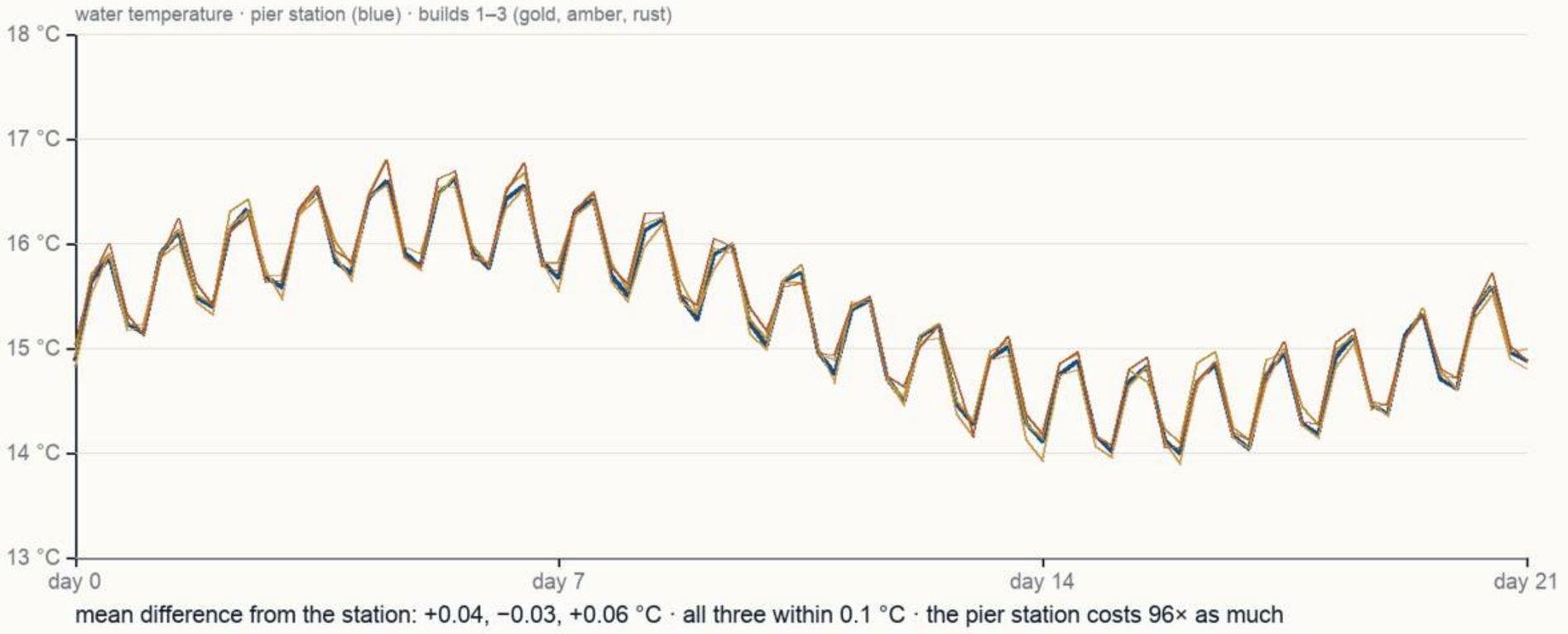
R4 radii on every inside corner of the cast caps (ENGR 310).

A bracket you can inspect from the pier without a diver, with the date written on it (ENGR 320, the one that rusted).

Tested to 10 m, five weeks in saltwater, three builds.



Three builds vs the pier, 21 days



mean difference from the station: +0.04, −0.03, +0.06 °C · all three within 0.1 °C · the pier station costs 96× as much

The thing that never failed is the thing you stopped checking

Build 2's bracket in week 3: the strap crept 2 cm and the housing began to knock the piling. The data showed it before the diver did: a 0.3 °C spike every swell.

The guide now has a step 11: write today's date on the bracket, and a date one month out. Everyone who built one has done it.

Three people who are not me

A marine science teacher at Point Loma High, with her class, in one afternoon. Their numbers matched the pier to 0.1 °C.

A lifeguard at Black's who wanted the water temperature on his own stretch. $38.90; he found a cheaper cell.

My mother, in the kitchen, from the guide alone, to prove the guide. It took her 70 minutes and it works.

WHAT'S NEXT

The kit ships 1 May.

sdit.org/eyes · the guide, the files, the data, all open · $44 as a kit, $0 as a PDF

The company that builds the forty-dollar sensor for the scientists who cannot buy the four-thousand-dollar one. The Portfolio is the pitch.